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SafeStreet: Empowering Women Against Street Harassment using a Privacy-Aware Location Based Application

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ABSTRACT

Sexual harassment of women in public places (e.g., footpaths, buses, and shopping malls) of major cities in developing countries is a growing concern. These harassments can happen in various forms ranging from commenting, catcalling, and staring to touching and groping, to attacking and raping. Though, the most severe form of harassments such as attacking and raping get some attention from the society, NGOs and law-enforcement agencies, unfortunately, other forms of harassments that are more widespread in public places remain largely un-attended or ignored in our conservative society. However, these harassments are more common and can have various negative psychological impacts on women that include a persistent feeling of insecurity, loss of self-esteem, restricted participation in daily life activities in public places. In this paper, we propose a crowd-powered privacy-aware location based mobile application, *SafeStreet*, that empowers women in public places against sexual harassments. *SafeStreet* allows a women to *privately* capture and share her own experiences in the street. *SafeStreet* enables a women to find a safe path, i.e., the path to a destination that has less harassment hazard, at any point of time.

Keywords

Sexual Harassment, Women Empowerment, Location Based Application

1. INTRODUCTION

Street harassment of women is a significant and prevalent problem around the globe. A recent study shows that more than 90 percent of women faced some sort of sexual harassments in public places [7]. Street harassment can happen in various forms ranging from commenting, catcalling,

and staring to touching and groping, to attacking and raping. Though, the most severe form of harassments such as attacking and rapping get some attention from the society, NGOs and law-enforcement agencies, unfortunately, other forms of harassments that are more widespread in public places remain largely un-attended or ignored in our conservative society. However, study shows that these harassments are more common and can have various negative psychological impacts on girls/women that include a persistent feeling of insecurity, loss of self-esteem, restricted participation in daily life activities in public places. Though the problem has been identified long before, no visible remedy or action can be seen to overcome this social problem. In this paper, we propose to utilize the power of GPS-enabled smartphones to empower women against street harassment, which in turn will facilitate women to feel secured, to boost self-esteem, and to participate in daily-life activities at public places.

Street harassments in a crowded urban area like Dhaka is a matter of great concern now-a-days. Women from all walks of life - from school/college going girls to working women face sexual harassments in public places - whether on streets/footpaths, or market places, or in public transports. Specially, in developing countries like Bangladesh or India, unsafe public transports, highly crowded streets/footpaths, lack of amenities or law forcing agencies, and male dominating social cultures, are the main reasons behind this widespread epidemic of sexual harassment in public places. Another reason for the increasing trends of sexual harassments of women in public places in conservative societies like Bangladesh, is that women remain silent in fear of shame and social disgrace against such crimes. A recent study in New Delhi found that 95% of the surveyed women believe that they had to restrict their mobility in fear of harassment in public places, where 82% of them faced harassments in public buses [7]. A recent study in Dhaka also shows that, public transport is the most favorable places for the predators, where as footpaths/streets and public gathering places are also major hunting ground for this type of crime [10].

Traditionally, mothers/older sisters shared tips and advices such as do not walk alone, or do not go out at night, or dress conservatively, to young girls to keep them safe by avoiding such incidents. However, at this stage of modern society where we all talk about equal rights for men and women, it is time to go beyond these advices and empower

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women in all means so that street harassment no-longer can limit a women to navigate public places to do their own jobs. Several recent efforts have been made to tackle this street harassment problem through the use of information and communication technology. The main theme of all these works are to allow victims to report the incident or call for immediate help to a friend by using their mobile phones [4, 3, 6, 8]. These approaches are certainly helping to raise awareness among women and in some cases empowering women to some extent. However, in most of the cases, the duration of the sexual harassment is very short (e.g., commenting/touching), thus a friend staying a bit further a cannot be of any help. More importantly, in a conservative societies like Bangladesh women do not want to report such incidents due to social shame and pressures.

To alleviate all the problems, in this paper, we propose a crowd-powered location based *mobile application*, *SafeStreet*, that empower women in public places against sexual harassments. The key idea of *SafeStreet* comes from a key observation that the most of the sexual harassments are *spatio-temporal* in nature, that is, a certain type of harassment occurs more frequently at a certain place at a particular time. For example, we have observed that un-wanted physical contact/touching is more frequent in Nilkhet area of Dhaka city between 6 pm and 8pm, whereas, commenting is more frequent in City College area in Dhaka city between 12pm to 2pm. *SafeStreet* continuously updates a user about the harassment risks at a place she is going to visit next. Moreover, *SafeStreet* allows a women to find a safe path, i.e., the path to a destination that has less harassment hazard, at any point of time. Again, for a casual traveler, who has the choice of visiting a place at a flexible time, *SafeStreet* suggests the time of visiting the place, when the harassment hazard is the least. Moreover, through *SafeStreet* law-enforcement agencies can take proactive actions such as deploying community police and volunteers in hot zone areas. Note that all the harassment data is spatio-temporally tagged, which is collected through crowd-sourcing and maintained in a location based server.

Another interesting and important feature of *SafeStreet* is that it allows a women to privately capture her harassment experiences on the go, by just taping the mobile screen twice. *SafeStreet* learns from these personal experiences and advice the user to take a safe path while traveling. This feature helps a women of conservative nature, who does not want the harassment case to be reported due to social shame. More importantly, *SafeStreet* allows a women to privately share their harassment information without disclosing their identities. In this privacy-aware approach, a women can form a circle with her close friends, and the system allows to report aggregate harassment incidents to the server so that individual's identity is not revealed to anyone.

In summary, in a country like Bangladesh with 114.808 millions of mobile phone subscribers [2], a mobile based application *SafeStreet* can play a vital role against street harassment. In this project, we propose to utilize the power of GPS-enabled smart phones which will help women to find a safe path with less harassment risk and empower women against street harassment. This in turn will facilitate women to feel secured, to boost self-esteem, and to participate in daily-life activities at public places.

2. LITERATURE REVIEW

Several recent efforts (e.g., [4, 5, 3]) have been made to tackle the street harassment against women by using information and communication technology. "Hollaback!" [4] is a social movement organization to raise awareness about street harassments against women. They primarily allow users to share their harassment stories through a web interface. These stories help users to know more about the types of harassments in different parts of the world. They have also developed a smartphone application for documenting (in words and pictures), mapping, and sharing incidents of any street harassment. In Toronto, a non-profit organization, Metropolitan Action Committee on Violence Against Women and Children (METRAC), launched a mobile app, "Not Your Baby" [5]. By using this app one can select the location and the type of harassment that she has experienced. Then the app pops up some supportive suggestions regarding how should she react to different situations of sexual harassment. "HarassMap" is a web-based platform in Egypt [3] that collects and summarizes different types of harassments in a map. By using HarrasMap one can report about any assault through a mobile phone or through a web interface. Recently, a web based system named "Protibadi" [8] is developed for sharing street harassment experiences in Bangladesh. "Protibadi" allows a user to report the location and nature of harassment, and to write a descriptive blog to share her own experience about sexual harassment.

Some other notable applications to combat street harassments are "Circle of 6", "bSafe", "Guardly" [9]. Circle of 6 is a mobile application that allows a female user to send an alert to her friends when she's in a threatening situation. In this app, a user can press the screen twice to send out one of three predetermined text messages to preferred contacts. bSafe sends an emergency message to one's chosen contacts with the push of a single button. Guardly is a personal safety service that instantly connects a person with her safety network and authorities in an emergency situation. It also places a phone call to contacts with victim's name, exact location and the type of emergency.

The main theme of all the above works is either to allow victims to report/share the incident using mobile phones or through web interfaces, or to enable them to call for immediate help to a friend by using their mobile phones. Though, these initiatives are raising awareness among women to some extent, users, in particular women in developing countries, are not keen to use these systems due to conservative nature of the society. In most of the cases, women remain silent against such crimes in fear of shame and social disgrace. According to a survey conducted by non-government organization ActionAid Bangladesh, 97% women feel that sexual harassment is a form of abuse but 81% of women subject to such abuse are scared to approach the police for help [1]. Also, in most the cases, the duration of the sexual harassment is very short (e.g., commenting/touching). Thus systems that allow a user to seek help from her friends become useless if her friends stay far away from the victim location. In this paper, we have developed a privacy-aware spatio-temporal location based application *SafeStreet* that allows users to share their experiences privately and receive continuous spatio-temporal alerts of possible harassment risk as they move on.

3. SAFESTREET: DESIGN AND DEVELOPMENT

3.1 Design Concept

3.1.1 Client-Server Model

SafeStreet works in a typical client-server model. In the server side, all the reported harassments are stored in a spatio-temporal manner. That is, for each incident - along with the harassment details, the location and the time of the incident are stored in a database. After receiving harassments records through crowd sourcing, i.e., reported by victims or observers or crime reporting agencies, the server summarizes these incident records occurred in different spatial resolutions at different time periods. On the other hand, the client side is an Android application. By using this application, a user can view the frequency of reported incidents of any region in a map, get the alert of possible harassments of the place where she is going to visit, or search for a safe route and a safe time before visiting a place.

3.1.2 Privacy Aware Reporting Protocol

One of the important features of SafeStreet is the privacy aware protocol that allows women to privately capture and share her own harassment experiences without revealing her identity to others. This privacy-aware feature of our application makes it different from all other available anti-harassment applications. For privacy-aware data capturing, we devise a protocol that allows a user to double-tap the mobile screen to automatically store the time and location of the harassment incident without making any public notice. Since a user might not want to share her harassment experience with others due to social shame, for privacy-aware harassment reporting, we propose a privacy-aware protocol that allows a user to share harassment experiences without revealing her identity. Note that hiding only a user's identity, i.e., name, IP address or phone number, cannot always protect a user's privacy because location and time together can act as an identifier for the user. For example, adversaries may have background knowledge about a user's location and time, or a user's location can be predicted by mobile signal triangulation. To overcome the above mentioned limitation, our approach makes a user k -anonymous [?] before revealing the user's harassment incident to the database.

In our approach, a user can make her own circle (or a group) with her friends or other users of the system. In a particular group, one group member is randomly selected as the coordinator of the group. When a user encounters a harassment, the harassment record is initially shared with the coordinator. The coordinator shares the harassment record with the server only when there are at least $k-1$ harassment records from other group members at nearby places and time. Before sharing k harassment records with the server, the coordinator replaces the location with a region that includes location of all places of harassment records. Similarly, the coordinator replaces the exact time of harassment records with a time range that includes time of k records. For example, if the user makes a circle with 15 members and among them 6 users have harassment reports (one from each user) to share, then total 6 reports can be uploaded in the system in a single pass. From the aggregated 6 reports, no one will be able to identify which one out of 6 reports belongs to the target user (here $k = 6$).

3.1.3 Safe Route

SafeStreet allows a user to find a safe route to a destination. If a user wants to travel from a source location to a destination location at a particular time, SafeStreet shows the safe path, as well as unsafe path to the destination based on the previously reported harassment incidents in that region. On the other hand, if a user is flexible to travel at anytime, SafeStreet shows the safe time and the safe path to a destination. In this case, the safe time refers to the time of the day when there is least risk of harassment on the path to the destination. In addition to finding the safe route, SafeStreet alerts a user in real-time about possible harassment risks at user's current place.

3.2 SafeStreet Prototype

As proof of our design concept we have developed an Android application (please refer to Figures 1 and 2). This application has four major features: visualizing harassments, capturing harassments, reporting harassments, and searching safe routes.

3.2.1 Visualizing Harassments

After logged in into SafeStreet mobile app, a user can see aggregated harassment reports of surrounding regions in a map, which were previously reported in the server (Figure 1(a)). The harassment statistics is shown by using hierarchical clusters, where a lower level cluster represents the frequency of reported incidents of a smaller area (e.g., roads) and an upper level cluster represents the frequency of reported incidents of a larger urban zone (e.g., city). A user can zoom-in or zoom-out the map to see the detailed or summary statistics, respectively.

3.2.2 Capturing Harassments

SafeStreet mobile app enables a user to capture any harassment experiences in real-time. In a typical scenario, if a user experiences or observes any harassment incident, the user can add the detail of the harassment by tapping on the map displayed on the home screen. However, a user may want to capture the harassment without noticing it publicly in fear of social shame. To facilitate the private capturing of the incident, a SafeStreet background service runs in the user's mobile phone and a user can just tap the mobile screen twice to capture the time and location of the incident. Thus, even if the incident occurs for a very short duration of time one can still capture the harassment record within few seconds. Later, the user can add the details of harassment records that the user encountered. Note that, a user can turn ON or OFF this background service using the Settings option of the application.

3.2.3 Reporting Harassments

By using SafeStreet mobile app, a user can upload the harassment records at any time to the server. However, as we have discussed, one may *not* want to publish the harassment experiences publicly in fear of social shame in a conservative society like Bangladesh. Thus, in our application, we incorporate privacy aware approach that allows a user to share the reports without revealing her identity. To anonymize the harassment reporting part, we allow a user to form a group/circle by adding her friends or any existing app users in her circle (Figure 1(b)), and then share the experiences as a group so that no one can identify the individual.

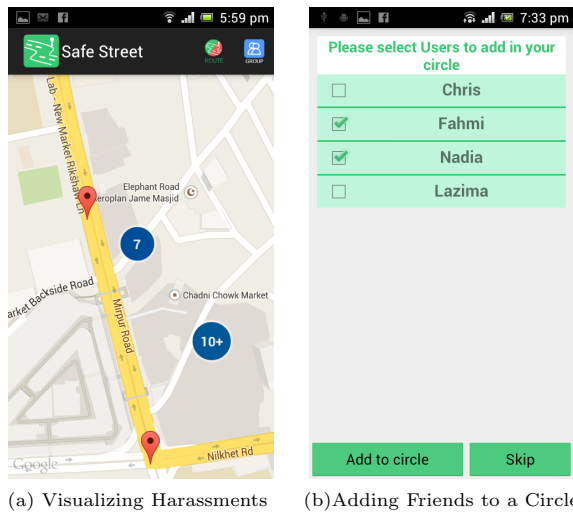


Figure 1: Screenshots of SafeStreet App

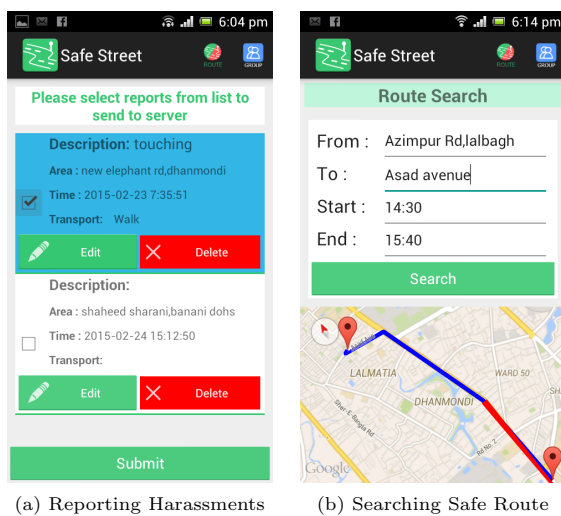


Figure 2: Screenshots of SafeStreet App

When a user submits a harassment report (Figure 2(a)), it goes to the coordinator of the group (a coordinator is a randomly chosen member in the group). When the coordinator receives a number of harassment records (say 6) from 6 users (say 3) in a spatial region at a certain time period, all these reports are uploaded to the server. Note that, from the Settings option of the application, a user can select whether she wants privacy-aware report sharing.

3.2.4 Searching Safe Route

In order to find a safe route, the application allows a user to give her starting location, destination and preferred time of travel, then if she clicks the 'Search' button then SafeStreet returns a safe path in her preferred time and shows the direction to the destination (Figure 2(b)). SafeStreet also shows the risky path if multiple paths are available to the destination, so that the user can avoid the path. Route search algorithm finds out the shortest path from the starting location of the user to her destination, considering the harassment records (location and time) to

be the weight. Figure 2(b) shows a route in a Google map. In this figure the red portion of the path means the algorithm has found harassment reports in this path and thus it is risky. The blue portion signifies the rest of the path is free of risk. Since the server maintains all accumulated harassment information, our server process runs the route search algorithm and returns necessary paths to the mobile user.

4. CONCLUSIONS AND FUTURE WORKS

We propose a privacy-aware location based mobile application, SafeStreet, that empowers women to combat sexual harassment in public places. Our application allows a user to capture on-street sexual harassment experiences privately. More importantly SafeStreet allows a user to share the data publicly in a privacy preserving manner, so that women in general can be benefited from each other. Our private sharing protocol allows a user to form a group with friends and share the data in an aggregate form to the public forum in a way that no individual's identity is revealed. We believe that SafeStreet will certainly help women to feel secured, to raise self-esteem, and to participate in daily-life activities at public places. In future, we will deploy the application in the field and conduct user studies to know user experiences about SafeStreet. Another future direction of this research is to devise appropriate methodologies to encourage users to contribute in the data collection phases.

5. ACKNOWLEDGEMENTS

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